

1 Introduction

This assignment asks you to continue adding to your network from the first assignment, which you had configured with an interior gateway protocol (OSPF), by adding an exterior gateway protocol (BGP) to connect your network to others. The aim of this assignment is that you configure and understand the operation of BGP on the Internet. This assignment gets you to configure BGP sessions with other ASes run by other students in your class on the mini-Internet to give you a feel for how the Internet works.

For guidance on configuring BGP, refer to Richard Sanger's mini-Internet project quick start guide [here](#). The official FRRouting documentation is also available [here](#): latest. As with the OSPF assignment, one of the skills you will learn is becoming familiar with searching through the documentation for answers.

1.1 Stages

This assignment guide will help you work towards the final configuration in a series of stages:

- i. Verify your OSPF configuration from assignment 1 is working correctly;
- ii. Add addresses to your external network links;
- iii. Configure iBGP sessions between your routers;
- iv. Configure eBGP sessions with external ASes;
- v. Advertise a route for your network prefix over BGP;
- vi. Configure BGP communities to share routes with peers on the IXP;
- vii. Configure *policy-based routing* based on whether a BGP neighbour is a peer, provider, or customer.

Each *stage* has one or more *steps*. The key to successful implementation and learning is to test each step carefully, so you're sure it works correctly and you understand what is happening before moving on to the next one. You will find a guide to commands that you can use to test what is happening at the Appendix A.

1.2 Assessment

This is a complex assignment and the work needs to start early; work in the last week alone will not be sufficient. The assignment will be assessed according to the final product and the history of development. Create a private project called “COMPX304-A2” in GitLab; do not delay the creation of this project.

Although a complex and time-consuming project, the “bar” for a pass mark (50) is relatively low (successfully complete everything up to and including step 9).

This assignment is an *individual* project, so work on your own. There are many ways of configuring the network such that it will work; develop your solution. You may discuss the assignment with others in general terms but do not look at anyone else’s configuration or show them yours. If you need help, ask us (lecturer and demos) or post a question on Moodle (but don’t post *any* configuration code on Moodle).

2 Network Topology

2.1 Your Internal Network Topology

Figure 1 shows the internal layout of your AS, which you configured with OSPF in Assignment 1. In Assignment 2, you need to configure six of the routers to use the links that connect them to routers in other ASes. You will configure BGP in such routers to connect your AS to ASes managed by other students in your class. The inter-domain connections are shown in blue, representing the business relationship. Figure 2 shows exactly which other ASes and IXP your AS is connected to.

2.2 The Internet Topology

Figure 2 shows the Internet topology. Routers ZURI, HAML, BOST, ATLA, PARI, and NEWY connect with another AS and are called “border routers”, while LOND and TRGA are internal routers. These other ASes are either customers, providers, or peers of your AS. The NEWY router is connected to an IXP (to a *route server* there), which allows your AS to exchange routes/traffic with multiple ASes.

The orange ASes are Tier-1 ASes (1, 2, 17, 18, etc.), meaning that all of their neighbours are either peers or customers. The grey ASes (15, 16, etc.) have no customers and in such cases are called *stub* ASes. We have pre-configured both the Tier-1 and stub ASes; their routers will use BGP to exchange routes with your AS (when neighbouring one of them).

You are managing a Tier-2 AS. Your AS is directly connected to two provider ASes (inwards the topology), two customer ASes (outwards the topology), and

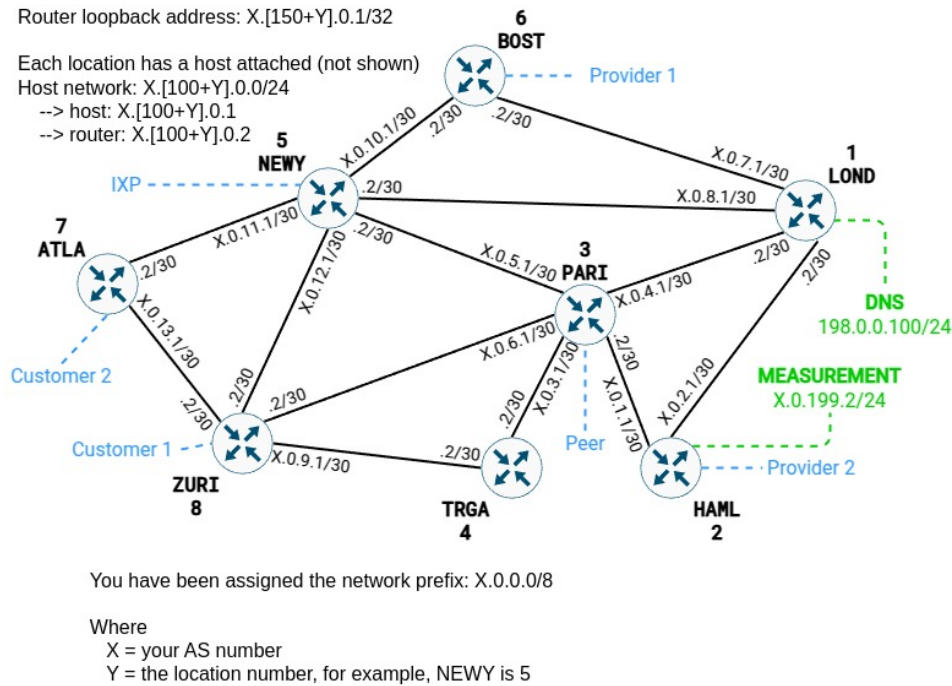


Figure 1: The internal layout and address plan of your AS. You have been assigned an entire /8 to number your network, X.0.0.0/8, where X is the network (autonomous system) number we have emailed to you. Each router shown has a host attached.

one peer AS (marked with a dashed, green line). In addition, your AS is a member of an IXP, through which your AS can connect/reach and peer with a good number of other ASes following the “non-paid peering model”. (Note that your AS will not use the IXP to exchange routes with customer and provider ASes.)

To illustrate, if you are AS 10, you are a customer of AS 7 and AS 8 (inwards), a provider for AS 11 and AS 12 (outwards), and a peer of AS 9 (same level). In addition, via IXP 122, you will peer with all the tier-2 ASes in the “opposite block” and which also are present in the same IXP. For these ASes, your AS will peer with and establish a peer-to-peer relationship (no money exchanged) with them. In the case of AS 10, it will have a peer-to-peer relationship with AS 19, AS 21, AS 23, AS 25, AS 27, and AS 29. Likewise, AS 92 would peer with AS 3, AS 5, AS 7, AS 9, AS 11, and AS 13.

The IXPs allow peering with multiple neighbours in one location and without paying a provider. Each IXP is running a BGP route server (RS) to advertise routes between all of its participants. For example, without IXPs, AS 10 would reach AS 23 via the (long) AS-path 10 8 6 4 2 17 19 21 23. However, once peered with IXP 122, AS 10 can reach AS 23 directly through the IXP.

Recall that BGP has two parts: iBGP (which connects all BGP routers inside the AS) and eBGP (which connects the border routers to border routers in other ASes). Initially, you will focus on configuring eBGP sessions with all of your neighbours to exchange routing information and iBGP sessions to prop-

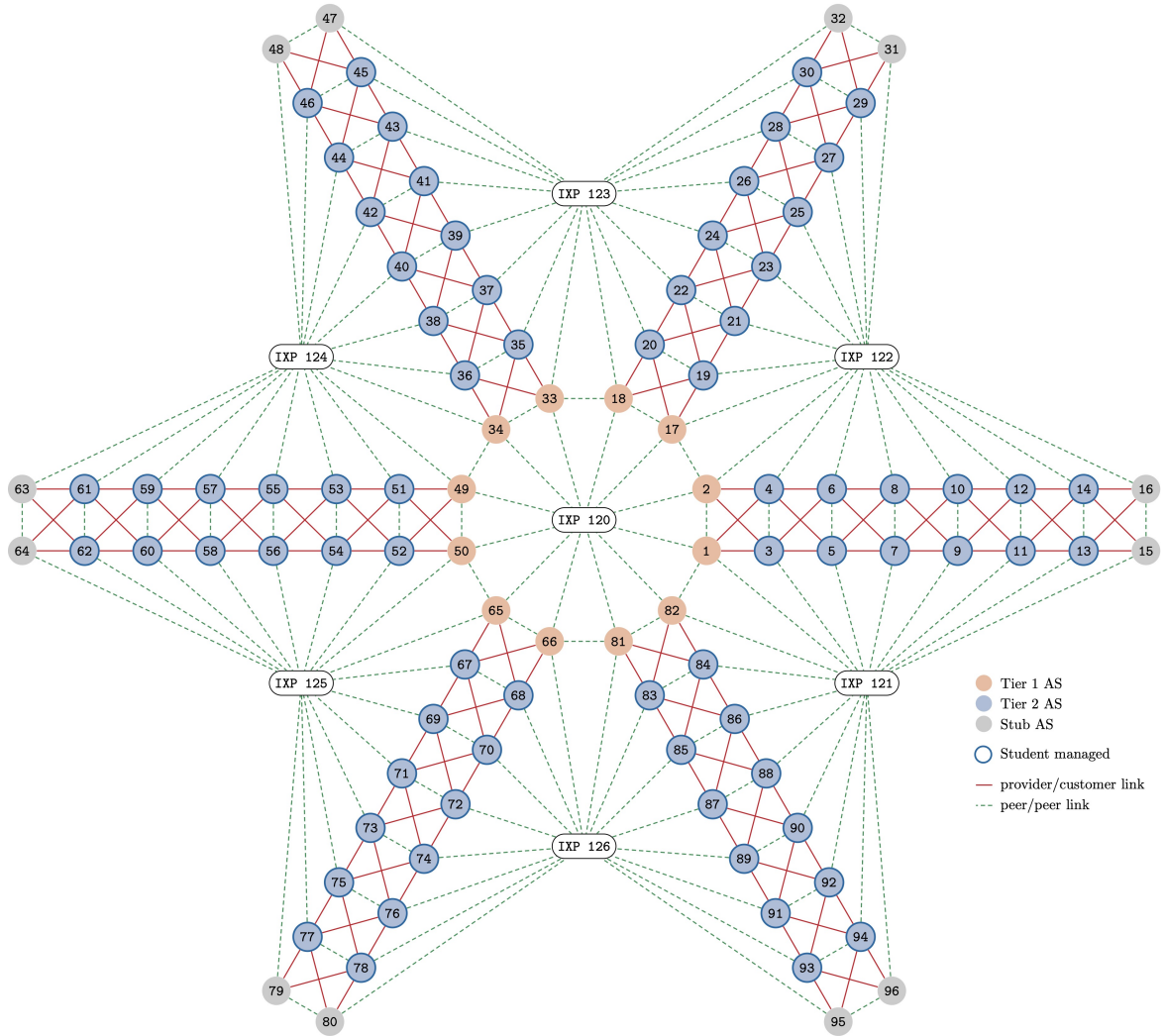


Figure 2: The topology of ASes on the mini-Internet. There are 12 Tier-1 ASes (in red), 12 Stub ASes (in grey), and 72 Tier-2 ASes. We have divided the Internet into 6 blocks, which are interconnected by tier-1 ASes and Internet eXchange Points (IXPs). Your AS has a connection to an IXP. We have assigned you one Tier-2 AS to configure, and we have pre-configured all of the Tier-1s, Stubs, and IXPs. Higher numbers in the same block, “outwards”, indicate customers (or customers of customers), while lower numbers in the same block (“inwards”) indicate providers (or providers of providers).

agate that information inside your AS. Later, you will need to configure BGP *policies* in border routers to determine which routes should be imported and exported.

2.3 Verifying Your Network Configuration

You can find suggestions of useful troubleshooting commands in the appendix at the end of this assignment. See Appendix A for useful FRRouting commands you can run. We have also provided several tools which you can use to verify your network configuration. These are very similar to the tools actually used on the Internet. You can find most of these tools linked on the main website for this project <https://mini.cms.waikato.ac.nz>.

2.3.1 BGP Looking Glass

A BGP Looking Glass (LG) service is a public, read-only portal that many network operators provide to expose public routing information to other network operators to assist with network debugging. This gives you insight into the BGP routes that another network can see. Refer to slide #32 “Looking Glasses”.

The Looking Glass service for the mini-Internet provides the BGP routing database for routers. It *periodically*¹ runs and saves the output of the command `show ip bgp` on the selected router of the selected AS. The LG is here: <https://mini.cms.waikato.ac.nz/looking-glass/>. By default, it shows a router (ZURI) of AS 1. You can select a different AS from the drop-down menu, and then a different router in that AS; this can be any AS, but it will be often your AS.

2.3.2 The Connectivity Matrix

You can track your own progress towards this assignment (and the class as a whole) using the connectivity matrix. The connectivity matrix is here <https://mini.cms.waikato.ac.nz/matrix>. The cells are coloured as follows:

- *green*: there is a working route between two ASes;
- *red*: the two networks are *not* reachable from each other;

As you and your classmates configure BGP, the matrix should turn gradually green as more of the Internet becomes reachable. After configuring OSPF in Assignment 1, your AS should have reachability to itself, i.e. the cell at row X and column X should be green. This is the diagonal green line down the middle of the matrix. NOTE: it may take over *10 minutes* for the matrix

¹Changes to the network may take a few minutes to appear in the LG; it should not be used to debug interactively!

to update fully, so your BGP configurations changes will *typically not appear immediately*.

The host that helps creating this matrix, called matrix host, is connected to the PARI router. It uses a pre-configured interface (**matrix_X**) with the IP address X.0.198.2/24, where X is your AS number. It pings to the ATLA host in the target AS to check reachability (it does so for all other AS on our Internet; imagine doing so for the real Internet!). In the previous assignment, you should have configured the ATLA host with the address X.107.0.1 and a default route that allows it to respond to the matrix host. If this was not properly configured, your AS will never turn green (which will be a sad experience).

2.3.3 Measurement Container

The mini-Internet has a measurement container that is connected to all Tier-2 networks, allowing you to ping or traceroute to any AS without crossing any other networks. This is a shared container, so *please be careful* not to change its configuration. Always close your ssh sessions cleanly, by typing **exit**.

You can access the measurement container from the proxy, using
`./goto.sh measurement`

You can launch a traceroute from any AS to a router or host in any other AS by using the `./launch_traceroute.sh` script. Use the “-i” argument (interface) and the target host or router. The container is connected to the HAML router of each AS.

For instance, to traceroute from AS 92 to the newy router in AS 20, use

```
traceroute -i group92 newy.group20
traceroute -i group92 host.newy.group20
```

It is possible to use ping as well, as below (note that the argument to indicate the interface to be used is -I).

```
ping -I group92 host.newy.group20
```

2.3.4 BGP Analyzer

The BGP Analyzer tool helps you check your BGP policy by letting you know of routes that you are advertising to peers or providers that you are paying for. BGP Analyzer periodically updates from the live routing tables, so it can take a while (5-10 mins) until it verifies a change. Do not rely solely on this tool; recall that a change in your neighbours’ policy may change the prefixes you advertise.

The BGP Analyzer can be used by selecting your AS in the looking glass tool and scrolling down to the bottom of the page, where you see “policy analysis”.

<https://mini.cms.waikato.ac.nz/looking-glass>

3 Configuration

This is where the actual assignment begins. Carefully follow the numbered steps to configure BGP on your network.

3.1 Keeping Notes

We recommend that you keep a very organised track of the commands that you use for each step in this assignment. You'll find you often will need to come back to them. And take note of any issues you have and submit along with your commits. Some steps require you to submit evidence; these are highlighted in **bold text**. Ensure that you **commit and push frequently**, preferably at every step.

3.2 Verify your OSPF Configuration

From the last assignment, you should have a working OSPF configuration providing full connectivity within your AS.

1. Verify that you have full connectivity within your AS and that the corresponding square (from your AS to your AS) in the connectivity matrix is green (<https://mini.cms.waikato.ac.nz/>). If not, correct your OSPF configuration so that your ATLA host can ping to X.0.198.1 and your square on the matrix turns green.

3.3 Configuring inter-domain interfaces

We have assigned the addresses for you to use on your inter-domain (external) links. You will find a list of all external links here <https://mini.cms.waikato.ac.nz/as-connections>.

You should use the filter to list only your connections. Once you have set the filter to your AS you should see 6 different links. On the left side will be the router in your AS that the link belongs to and your IP address that you will assign to the interface; on the right side is the AS you will be connecting to with the IP address that they are assigned. Above each connection listed here is the relationship of the connection. If above your interface it says customer, that means that you are the customer and you are connecting to one of your providers. You have 2 connections where you are the customer, two links where you are the provider, one link to a peer and one link to an IXP.

The interfaces for inter-domain links that you will be configuring are named `ext_<external AS>_<external router>`. However, you can also check on the router (`show interface brief`) and look for the `ext_*` interface. In contrast, router NEWY is always connected to an IXP and its interface is instead

named `ixp-<AS>` e.g. `ixp_126`. These links match with our Internet topology, shown in Figure 2, and your internal topology, shown in Figure 1.

2. Following the guidance above to find your external connections and IP addresses, assign an IP address to each of the interfaces with an inter-domain link (external). NOTE: each router has a max of *one* external connections with TRGA and LOND not having any external connections. This makes for a total of six external connections to assign IP addresses to (do not overdo it!).

Save the configuration, copy to the jump host, uncompress it, and make sure all files are in the repo. Commit the changes and push.

3.4 Start BGP configuration in the router

3. At each router, start configuring BGP (this starts a BGP daemon) by informing your ASN. Then, assign the BGP router's id (which will be the same of the OSPF router id). Check the BGP router configuration section of the FRRouting manual (here).

Save the configuration, copy to the jump host, uncompress it, and make sure all files are in the repo. Commit the changes and push.

3.5 Establishing iBGP Sessions

In order to become familiar with configuring BGP sessions, start by configuring iBGP sessions between your routers. As you control both routers in an iBGP session, it is easier to debug and see what is going on. By default, iBGP only shares routing information received via an eBGP session. NOTE: until you configure BGP to share your network prefix (your X.0.0.0/8), these iBGP sessions will *not* share any routing information. You will share your network prefix later in step 15. **So, for the time being, you will see no actual effect apart from being able to see the connections between your routers.**

4. Configure internal BGP sessions between all of your routers. You will need to establish a *full-mesh* in which every router has an iBGP session (a TCP connection) with every other router in your AS. Therefore, you need to configure each of the 8 routers (in a similar way).

You will use the command `neighbor PEER remote-as ASN`, as shown in the manual. The information needed to establish each iBGP session:

- (a) the loopback IP address of the router you are connecting to;
- (b) your ASN again (the ASN of what could be a remote AS, but for iBGP, this is again your own AS);

5. Configure the `update-source` for each iBGP session in each router to use the loopback interface (10). If you fail to set this, BGP will connect by default from the address on the “closest” physical interface, which might cause a problem depending on the order you take the steps. The remote router needs to be configured to use this interface; if the IP addresses are different, BGP would not establish a session.) Use the command `neighbor PEER update-source <IFNAME|ADDRESS>`, as in the manual.

NOTE: at this step, you are *not yet* sharing any routing information over BGP! You have simply established the iBGP sessions.

6. Verify your iBGP neighbours are connected by running
`show ip bgp summary`
Also, look at the routes being shared via BGP by running
`show ip bgp`

Is this what you expected?

Note this in a text file called `step-6.txt` and add it to the repo along with the current configuration.

Save the configuration, copy to the jump host, and make sure all files are in the repo (including notes). Commit the changes and push.

Completing correctly up to this point is worth 30 marks.

3.6 Establishing eBGP Sessions

In this section, you will configure your eBGP sessions, starting with the IXP. You configure these in much the same way as iBGP, but you will specify a different “remote AS”. NOTE: because you are establishing these connections with others in your class, they might not have configured their router yet, so the session might not be established yet, despite you doing everything correctly.

However, you should be able to establish the eBGP session with the IXP route server immediately, so *configure this first* and ensure it is working before you configure your other eBGP sessions. Again, thus far, we are simply establishing BGP sessions without sharing our prefix (X.0.0.0/8) yet.

7. In this step, we will configure NEWY to establish an eBGP session with an IXP route server. Which IXP? IXPs are assigned ASNs from 121 to 126; you are connected to (*only*) one of these IXPs. You can verify the IXP you are connected to by checking Figure 2 (or at <https://mini.cms.waikato.ac.nz/as-connections>, selecting your ASN and looking for the IXP on the list). NEWY will connect to the router server in your IXP via the address 180.X.0.X, where X is your IXP’s AS number as usual.

You will need to have:

- (i) the IP address of the remote router; for eBGP, we use the *interface address* rather than a loopback address;

- (ii) the ASN of the remote network (e.g. the IXP or another neighbour AS);
- (iii) the local IP address that is used for the connection; this is known as the `update-source`. For eBGP, `update-source` should be the address of the (local) interface used with the external link (e.g.). (By default, FRR will use the closest physically connected interface and will likely pick to correct one. It is always best practice to set this explicitly to avoid any surprises.)
- (iv) a `route-map` to *permit* all routes.

The first step is to configure BGP in NEWY the information about the IXP neighbor and `update-source`, where IXP is the ASN of your IXP and X is your ASN.

```
neighbor 180.IXP.0.IXP remote-as IXP
neighbor 180.IXP.0.IXP update-source 180.IXP.0.X
```

The addresses `180.IXP.0.IXP` and `180.IXP.0.X` represent the different ends of the connection between your AS and the IXP (route server). While `180.IXP.0.IXP` corresponds to interface at the IXP and is common to all ASNs that connect to the IXP, the address `180.IXP.0.X` is *your* local interface that will connect to the IXP.

The next step will guide you to create a route-map.

8. FRRouting follows RFC 4271, which states that an eBGP session should not accept or advertise routes without an explicit route-map defined. This does *not* apply to iBGP, which will accept and advertise all routes by default. For now, add a “permissive” default route-map that will accept and advertise all routes (BGP policy-agnostic). This will allow us to import all routes announced by the peers on our IXP as well as to export our routes to the members in the IXP. This is *not* good, but we will fix this later (in Step 19).

You will use the command `neighbor PEER route-map NAME [in|out]`, as indicated in the manual. Follow this example (considering AS 99) to install a route-map that allows all routes; replace IXP with the ASN of your IXP, e.g. 122. The empty `ALLOW_ALL permit 99` rule always matches and permits all routes.

```
NEWY_router# configure terminal
NEWY_router(config)# route-map ALLOW_ALL permit 99
NEWY_router(config-route-map)# exit
NEWY_router(config)# router bgp 99
NEWY_router(config-router)# neighbor 180.IXP.0.IXP
→ route-map ALLOW_ALL in
NEWY_router(config-router)# neighbor 180.IXP.0.IXP
→ route-map ALLOW_ALL out
```

9. Verify the eBGP connection with the IXP is established using `show ip bgp summary`

Also, take a look at NEWY's BGP routing table using
`show ip bgp`

Does this show what you expect? Remember this shows the BGP routes you are *accepting* from your neighbours and not the routes you are exporting to them. Note the results of these commands in a text file called `step-9.txt` and add it to the repo along with the current configuration. **Save the configuration, copy to the jump host, uncompress it, and make sure all files are in the repo (including notes). Commit the changes and push.**

Completing correctly up to this point is worth 50 marks.

At this point, you have still not advertised routes for your own network into BGP. Let's fix that now. The easiest way to advertise our routes into BGP is to simply advertise *all* of our OSPF routes into BGP, by using `redistribute` to redistribute routes. Advertising OSPF routes into BGP is not recommended on a real network, as it fills BGP with many small prefixes rather than a single summary prefix for your entire network (this pollutes tables). Regardless, it is an easy way to advertise our network; we will fix this later.

10. Configure *only* the router NEWY to `redistribute` OSPF routes into BGP. NEWY is connected to all other routers in your AS; this will propagate over iBGP the OSPF routes to all of your routers. Do not redistribute OSPF on any other routers. You will remove this later.
11. Verify that you are exporting and importing BGP routes. Check that your iBGP mesh is carrying the routes learnt from the IXP via eBGP to the other routers on your network. Do so by checking their routing tables (e.g. run `show ip bgp` at the LOND router and see if there are one more routes in which the next hop is the IXP).

Conversely, check that the IXP has imported your routes by checking the looking glass website. You should see the routes to your networks. NOTE: the IXP will *not* advertise your routes to other AS members in the IXP until the BGP communities are added to the announcements, which is only done later in step 19 .

12. Following the example for the IXP, configure one eBGP session to each of your 6 neighbouring ASes. You should have 5 more eBGP sessions to establish, consisting of 2 customer links, 2 provider links, and 1 peer (alongside peering with the IXP). Refer to the webpage <https://mini.cms.waikato.ac.nz/as-connections> and Figures 1 and 2 to find your neighbouring ASes.

You need to establish your eBGP session with the IP address (part of a /30) on the other end of the external, inter-domain link. Use the command `neighbor`, as before.

For example, if you are AS 10, from here, you would have configured 179.0.65.2/30 locally on HAML interface `ext_7_ATLA`. This means the

remote IP address for the eBGP session is 179.0.65.1 and the remote AS is 7. If the neighbour is a Tier-1 or Stub AS, the BGP session should establish right away as these are pre-configured. Otherwise, eBGP sessions to Tier-2 neighbours will only establish if your classmate has successfully completed this same step on their AS. Note: you will not lose marks if you cannot establish a connection because your neighbour has not configured their AS.

13. Verify your eBGP sessions are established using `show ip bgp summary` in the routers that have external links. Also, verify that the routes learnt from eBGP are being shared via iBGP to all of your routers by checking the routing table (recall that in an earlier step we established a mesh of iBGP connections among routers). Refer to Appendix A.3 for more information about FRR commands.

Save the configuration, commit the changes and push to the repo.

Correctly configuring the network up to this point guarantees 70 marks.

3.7 Advertising Your Network Prefix to BGP

Thus far, you have established all the BGP sessions you will need. But you are advertising your network to the Internet as many small /30 and /24 prefixes. This is inefficient and means all other routers on the Internet need to carry these routes (you are inflating all the routing tables). So, next, you will replace your BGP advertised prefixes with a single /8 network prefix for your entire network.

At this stage, we also need to create a “blackhole route” for our /8 prefix, because FRR will not advertise a route to BGP that does not exist in its routing table. A blackhole route is a way to tell the router that when it needs to forward packets that match this prefix, they should be *discarded*. At first sight, this may sound like a really bad idea, but remember that *longest prefixes match first*. So, a blackhole for your whole /8 prefix will only match packets destined for parts of your network that you do *not* have more specific (/9 or longer) routes. In other words, it will discard packets addressed to those parts of the network that do not exist. Additionally, if you don’t add a blackhole route for your prefix, when the router receives a packet for an address in a subnet that is not used in your network, it could send it to another router advertising the /8 prefix. At best, this is wasteful and, at worst, it creates a routing loop!

14. Now it is the time to remove the redistribution of OSPF routes into BGP (which happens at NEWY router). You can reverse FRR configuration by putting a `no` before the command.
15. Configure BGP to advertise your /8 prefix. When you announce your /8 from one router, iBGP will carry your prefix to all other routers in your AS. This works, but for resilience, you should advertise your /8 prefix

from *all* of your routers. This way you avoid the router advertising your prefix becoming a single point of failure.

NOTE: a further step is required to advertise our prefix through the IXP. The IXP uses BGP communities to filter which peers it advertises routes to. You will address that later, in step 19.

16. On each router, use the command **network** to add a blackhole route for your /8 prefix.
17. Verify your configuration has worked. Use **show ip route** to check that the blackhole route is showing correctly in the routing table.
18. (Wait some 10 minutes and then) check the connectivity matrix to see which networks have become reachable. The matrix takes some time to update. You might also want to try running pings from a **host** to the router's loopback address (such as AS.151.0.1) on ASes that are connected with your AS, if any. It is best to ping from a host (pinging from a router might use the source IP address from an inter-domain link that you are not advertising over BGP, and it may be unreachable by the remote AS). My router may be exporting my route correctly, but how do I know if my neighbour has imported my route? You can use the Looking Glass (Section 2.3.1) to verify your prefix is in your neighbour's route table.

Save, commit and push the configuration.

Completing correctly up to this point is worth 80 marks.

3.8 Advertising your BGP Prefix to IXP Peers

The IXPs are configured to filter which peers they announce a BGP prefix to based on the BGP community value associated with the route. Therefore, you need to specify BGP community values on the route advertisements you send to the IXP. As discussed in the lectures, a BGP community value (of default type) is comprised by two 16-bit values separated by a colon (in some communities, because ASNs are now 32 bit, the coding is different). A route can have multiple BGP community values associated with it (like multiple “tags”). To advertise your network prefix through the IXP with the AS number IXP to the peer with the AS number X, you must add the community value IXP:X.

Check the examples presented in the lecture and visit the links (to the right place in the FRR manual) available on the slides. Also, refer to the mini-Internet tutorial² on **route-maps** to help you with this step. Take a moment to read the tutorial before attempting the next steps.

19. On NEWY router, create a route-map named IXP_out that *matches all routes* and *sets* a list of BGP communities that tell the route server to

²https://github.com/rsanger/mini_internet_project/wiki/2.5.6-Configure-BGP-policies

advertise the routes to your IXP peers. You should only advertise to your peers in the “opposite block” to your own (opposite may be 90 degrees), as within your “block”, you have customer+provider relationships to honour. For example, if you are AS 10, you should set the community values 122:17, 122:19, 122:21, 122:23, 122:25, 122:27, and 122:29, 122:31. AS 10 should *not* announce to AS 8 and any other in the same block via the IXP.

20. Now you have created the route-map you need to apply it to your BGP session with the IXP. Apply the route-map in the `out` direction to apply it to all routes that you advertise to the IXP.

21. Check that you are advertising your prefix to the stub and Tier-1 ASes (and any other BGP sessions which you have established) in the opposing block. You can use the looking glass, connectivity matrix, and ping the loopback at their (single) router to confirm.

For example, from TRGA host it should be possible to ping `tier1.151.0.1`, where `tier1` is the ASN of your tier-1 provider. For AS 10, it is ASN 17, therefore `17.151.0.1`.

22. **Save, commit, and push your changes, naming the step as usual.**

Completing correctly up to this point is worth 90 marks.

3.9 Applying Policy

Currently, you are not filtering the routes which you are advertising through BGP. By not filtering routes, you will be paying to carry traffic for networks that are not paying you (i.e. your peers) over links that cost you money (i.e. your providers). Remember that in BGP routing decisions are often made based on financial relationships rather than the best path.

The goal of this section is to only advertise your network prefix and customer routes to all of your BGP neighbours. You must advertise to your customers *all* routes you learnt from the Internet. However, do not advertise routes learnt from peers or providers to each other, including between two providers or peers. This would provide your peer free transit using your upstream which you are paying for. Recall the route leaks discussed in the lectures. Note that the ASes connected through the IXP are all considered peers. You can use the BGP Analyzer (§ 2.3.4) to help.

One way to apply this policy is to tag incoming routes with an “informational” BGP community value indicating if they are a customer, peer, or provider (check BGP communities in the slides). Then, filter outgoing routes based on where they were learnt from (e.g. do not announce to peer a route learnt from another peer).

To implement this policy, you will need to use `route-maps` and BGP communities. These new instances of `route-map` (e.g. I called them `FROM_PROVIDER`, `TO_PEER`, etc.) will replace your `ALLOW_ALL` and `IXP_out` route-maps, so ensure

they also include that existing behaviour such as advertising communities to the IXP. Essentially, in importing routes, you will match the BGP communities according to the relationship, while exporting, you will set the community. Your community numbers will be $X:Y$, where X is your ASN, and Y represents a code, an arbitrary value, that *your* define and use (you need three different values, to represent customer, peer and provider).

23. Apply BGP policies so that you only carry traffic through your network on behalf of your customers. But, don't carry traffic through your network on behalf of your peers or providers as this just costs you money.

Include in the repo a text file with a short report named `bgp-policy.txt` with a brief description of how you applied this policy. Include a before and after snapshot of `show ip bgp neighbors <address> advertised-routes` (or a looking glass snapshot) that shows that your configuration has worked.

24. You are done. Look over your configuration by using the `show running-config` command and clean up anything extra. Remember that you can use the `no` command to delete blocks of configuration.

Save, commit, and push the final version in the repo. We will assess the latest version committed before the time we start marking the assignments (we will let you know when it happens.).

4 Assessment

This assignment is assessed entirely on success and correctness. Though you will not be penalised for not being able to connect to a neighbour if they have not configured their network. The marks will be allocated as shown in table 1.

Steps	Description	Marks
1 – 2	Interfaces and OSPF configured	1
4 – 6	Established iBGP sessions	2
7 – 13	Established eBGP sessions	2
15 – 17	Advertise your prefix over BGP	3
19 – 22	IXP communities	1
23 – 24	Policy-based routing	1
		10

Table 1: Distribution of Marks

Your assignment must be submitted electronically using GitLab, like A1, with the name “COMPX304-A2”.

Your project in GitLab will include all the host and router configuration files saved as well as the text notes as requested in some of the steps. The history of commits must demonstrate the gradual development of the work, with the evolution.

Appendices

A Testing and Troubleshooting

A.1 Useful FRRouting Debugging Commands

Troubleshooting in FRRouting is run from enable mode (not configuration mode) most often using commands beginning with **show**.

? View the commands available to you, in any mode

<tab> Try and auto-complete a command, will list known options like interfaces names or addresses etc.

show running-config Show your configuration

show ip route Show all routes learnt from any source

show ip bgp Show all routes learnt from BGP

show ip ospf Show all routes learnt from OSPF

show ip bgp summary Shows the status of your BGP neighbours

show ip bgp neighbors <address> routes Show the routes received, after route-map filtering, from this neighbour

show ip bgp neighbors <address> advertised-routes Show the routes you are advertising to this neighbour, after route-map filtering

show ip bgp route-map <name> shows the result of applying this route-map to your routing table, i.e. what this route-map would advertise.

ping + traceroute are simpler versions of Linux tools.

A.2 Forwarding and Route Information

show ip route. The command (from enable mode) **show ip route** shows the routers forwarding table. This includes the routes that have been selected from the various routing protocols, static routes and directly connected networks. The selection is done, in part, using the *administrative distance* which is a priority. Note that the source of the route (where learned from) is included in the output.

A.3 BGP Information

- `show ip bgp summary`: shows what neighbours BGP is communicating with and statistics about the connection. It is a good quick check if something seems not to be working with BGP.
- `show ip bgp neighbors`: shows more detail than the previous command.
- `show ip bgp summary`: check the number in the **State/PfxRcd** column; if the column shows a state of **Idle** then you likely have a mismatch with your neighbour. If you see **Active** if your neighbour has not configured BGP yet.

show ip bgp. Probably the most important BGP information is given by the command below, which shows the list of BGP paths that the route is seeing, including the AS path. For example the entry:

```
show ip bgp
...
* 172.16.128.0/21 192.168.127.246          0 64999 65001 i
*>                10.254.254.253          0 65111 65001 i
```

means that BGP knows two ways to reach the 172.16.128.0/21 subnet. One is via AS65001 then AS64999, the other is via AS65001 then AS65111. The route chosen by BGP is marked with > at the left, which in this case is the second one.

The command `show ip bgp regexp 65111` will show just those paths that include AS65111.

A.4 OSPF Information

A few useful commands:

- `show ip ospf`: prints information about the OSPF area;
- `show ip ospf neighbor`: shows what neighbours OSPF is communicating with. It is a good quick check if something seems not to be working with OSPF.
- `show ip ospf interface`: shows the instance of OSPF that is communicating over the subnet attached to that interface. It includes, for example, the number of neighbours found and the identity of the DR and BDR.
- `show ip ospf route` and `show ip ospf database`: show the information that OSPF knows about the network, including all the prefixes that it has learned.